
Interactive session

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Introduction

When working in an interactive session a defined set of HPC (cloud) resources are allocated to you. Consequently, you cannot (by mistake) take over a whole login node by running a very heavy tool or command. Contrary to when sending a job to the queue system as a batch job, you will see error messages and standard output of your commands/runs directly in the terminal.

An interactive session is a good way of trying out your code before submitting your script/commands as a [batch job using qsub](#) (link only accessible from the NGC Wiki).

Note

If you run compute heavy tools and commands directly from one of the login nodes these can be subjected to enforced shut down.

User creation and modification

As a user on the NGC-HPC or an NGC-hosted self-managed cloud you have access to interactive

sessions by default.

Logging into the system for the first time

No special requirements.

Username and password

There is no password restriction on this solution as long as you are logged in to Citrix-NGC (<https://secureremote.ngc.dk>) and NGC-HPC or VMWare Horizon and a Self-manged cloud, you will have access to running interactive sessions (see details below).

How to use an interactive session

1. Connect to a login node

From Citrix-NGC

This solution is for diagnostic (clinical services) users and research users on **NGC-managed cloud**.

First you need to log in to Citrix-NGC (<https://secureremote.ngc.dk>). [Here](#) (link only accessible from the NGC Wiki) is a guide on how to log in to Citrix-NGC.

Then you need to access NGC-HPC.

Alternative 1) Open the MobaXterm app and log in to NGC-HPC through one of the login nodes.

Alternative 2) Open NGC Desktop, then open MobaXterm from the desktop icon and log in to NGC-HPC through one of the login nodes.

See this guide for more details on how to complete the above steps: [NGC Desktop and NGC-HPC](#) (link only accessible from the NGC Wiki)

From Self-managed cloud

This solution is for research users with access to **Self-managed cloud**.

Open VMware Horizon Client (<https://spc.ngc.dk> is the connection server).

Enter your NGC username and password and click **Login**.

You will receive a notification on your phone from Entrust Identity app. Click on the app notification, login with your chosen PIN (or face recognition) and choose **Confirm**.

You are now logged into VMware Horizon client and can see an overview of the services you have access to.

Double click on COS-CLDXXX to login to a CentOS desktop.

Open the Terminal (double click the Terminal icon on the desktop) and ssh into a login node. Enter your password.

```
# ssh into a Login node
ssh -X <username>@cld<cloud_id>-0001
# example with user jondoe on cloud 099
ssh -X jondoe@cld099-0001
```

The **-X** option makes it possible to open graphical user interface tools like e.g. Firefox from your session.

2. Start an interactive session

Using "iqsub"

This solution is only available from NGC-managed cloud.

In MobaXterm you can start an interactive session by using iqsub.

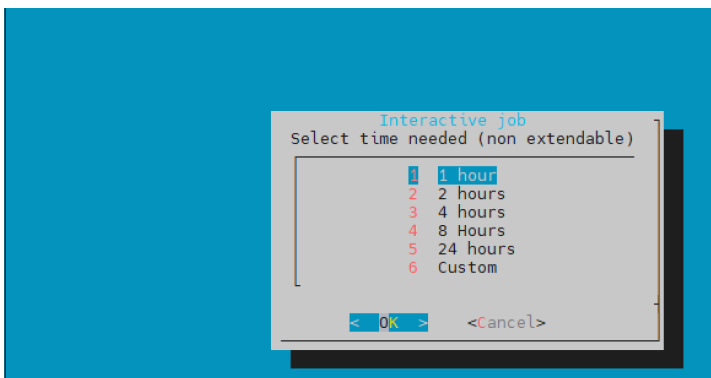
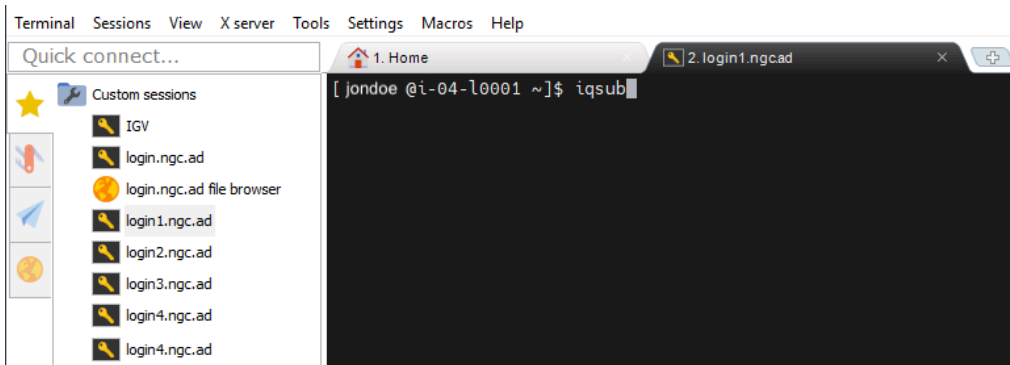
From one of the login nodes write **iqsub** and press **Enter**.

```
iqsub
```

First you need to specify which group you belong to. Then you can specify which resources you want (hours, cpu (number of processors), memory, GPU support). If you are not sure which resources you need, just use default values. One value to be aware of is the time needed.

Note

Your interactive session will close down after the time chosen and a running analysis will be aborted. The default value is one hour.



Using "qsub"

You can also start an interactive session using the qsub command, with the option **-I** for interactive session, and specify resources as below.

From one of the login nodes write "qsub -I -X -W group_list=<group name> -A <group name> -l nodes=1:ppn=4,mem=20g,walltime=02:00:00", exchanging <group_name> for your own group name.

```
# example: interactive session for a user in group named ngc-bio, 1 node and 4
CPUs, 20 GB memory and running for 2 hours
qsub -I -X -W group_list=ngc-bio -A ngc-bio -l
nodes=1:ppn=4,mem=20gb,walltime=02:00:00
```

The **-X** option makes it possible to open graphical user interface programs like e.g. Firefox.

To give more memory to your session you can change "mem=20gb" to "mem=100gb".

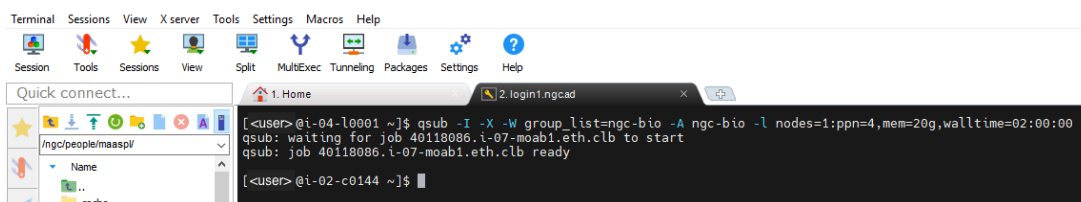
Walltime is specified in hours, minutes and seconds likes this hh:mm:ss, e.g.

"walltime=48:00:00" demands a 48 hour session. If you write walltime=48 this corresponds to 48 seconds.

You may have to wait for a little while before your interactive session starts. When it has started it will say "ready" and you can again enter commands, now in an interactive session.

To exit your interactive session write "exit" and press **Enter**.

```
# exit interactive session
exit
```



If you do not know (or have forgotten) your department's or group's name on NGC-HPC or your NGC-hosted self-managed cloud, you can write "id" in the terminal to see info about your user, including the group(s) you belong to.

```
# see user information  
id
```

Conclusion

You should now be able to run interactive sessions on NGC-HPC or and NGC-hosted self-managed cloud.

Relevant links

How to submit a job to the queue system: [Submitting jobs to the queue system](#) (link only accessible from the NGC Wiki)

Error handling

Nothing relevant.

Support

You can get help with issues related to the NGC services by writing an e-mail to support@ngc.dk. You can significantly speed up the support process by providing complete and detailed information for faster troubleshooting and error correction.

In the email to support@ngc.dk please include the following information (as detailed as you can in relation to your knowledge):

- An informative e-mail subject line.
- Your name, NGC username, and the [project \(user group\)](#) you belong to.

- When did the problem occur, how often does it occur and duration of issue.
- Which cluster? Which node? Which job?
- What happened? What exactly were you doing/trying to do? Include information on which commands you were using and/or which tools/modules (incl. versions).